lab12.py - C:/Users/Deekshith/AppData/Local/Programs/Python/Python314/lab12.py (3.14.2)

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```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("C:\\Users\\Deekshith\\Downloads\\12laptops - 12laptops.csv")
print("Dataset:")
print(df)

# Matplotlib Boxplot - Screen
print("\nBoxplot for Screen Size")

plt.boxplot(df["Screen"], notch=True, vert=False, showmeans=True)
plt.xlabel("Screen")
plt.title("Boxplot - Screen Size")
plt.show()


# Matplotlib Boxplot - Weight
# ------------------------------
print("\nBoxplot for Weight")

plt.boxplot(df["Weight"])
plt.ylabel("Weight")
plt.title("Boxplot - Weight")
plt.show()


# Seaborn Boxplot
# ------------------------------
print("\nSeaborn Boxplot for Weight")

sns.boxplot(x=df["Weight"])
plt.title("Seaborn Boxplot - Weight")
```

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```python
print("\nSeaborn Boxplot for Weight")

sns.boxplot(x=df["Weight"])
plt.title("Seaborn Boxplot - Weight")
plt.show()

print("\nSeaborn Boxplot - Weight vs RAM")

sns.boxplot(x=df["Weight"], y=df["RAM"])
plt.title("Weight vs RAM")
plt.show()


# IQR Method
# ------------------------------
print("\nIQR Method Calculation")

Q1 = df["Weight"].quantile(0.25)
Q3 = df["Weight"].quantile(0.75)
print("Q1:", Q1)
print("Q3:", Q3)

IQR = Q3 - Q1
print("IQR:", IQR)

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
print("Lower Bound:", lower)
print("Upper Bound:", upper)

# Outliers
outliers = df[(df["Weight"] < lower) | (df["Weight"] > upper)]
print("\nOutliers in Weight:")
print(outliers)
```

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```python
# Matplotlib
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("C:\\Users\\Deekshith\\Downloads\\12laptops - 12laptops.csv")

plt.boxplot(df["Screen"], notch=True, vert=False, showmeans=True)  # notch(curve) around the median
plt.xlabel("Screen")
plt.show()

plt.boxplot(df["Weight"])
plt.ylabel("Weight")
plt.show()

# Seaborn
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df["Weight"])
plt.show()

sns.boxplot(x=df["Weight"], y=df["RAM"])
plt.show()


# IQR Method   Outlier is a value that lies outside the IQR range; here weight 4.5 exceeds the upper bound (4.25), so it is an outlier.
Q1 = df["Weight"].quantile(0.25)
Q3 = df["Weight"].quantile(0.75)
IQR = Q3 - Q1  # Interquartile Range = Q3-Q1 (formula)
lower = Q1 - 1.5 * IQR    # standard statistical rule (Tukey's rule)-normal outlier, extreme outlier-3
upper = Q3 + 1.5 * IQR
outliers = df[(df["Weight"] < lower) | (df["Weight"] > upper)]
print(outliers)
```

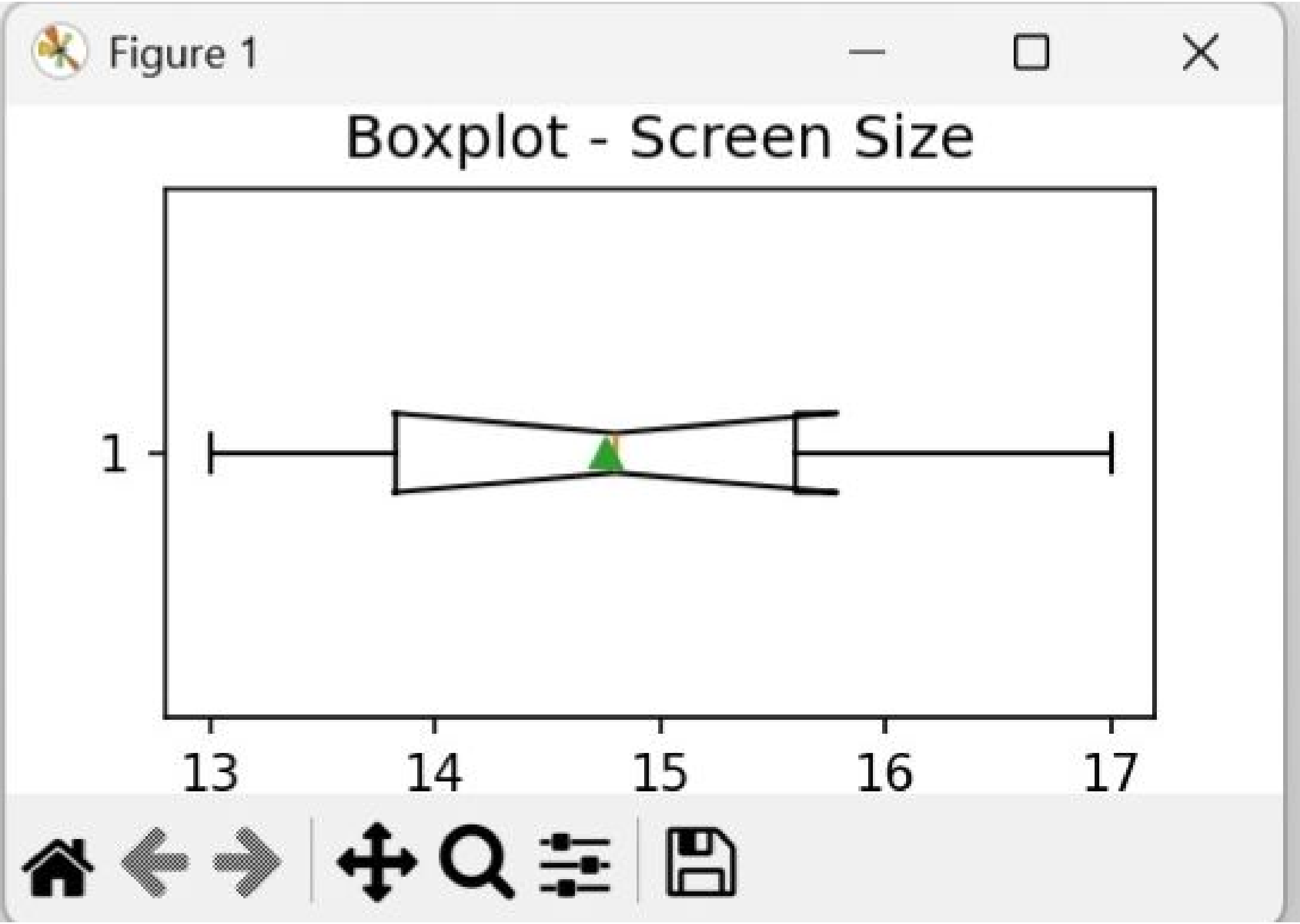
Figure 1
Boxplot - Screen Size
1
13
14
15
16
17

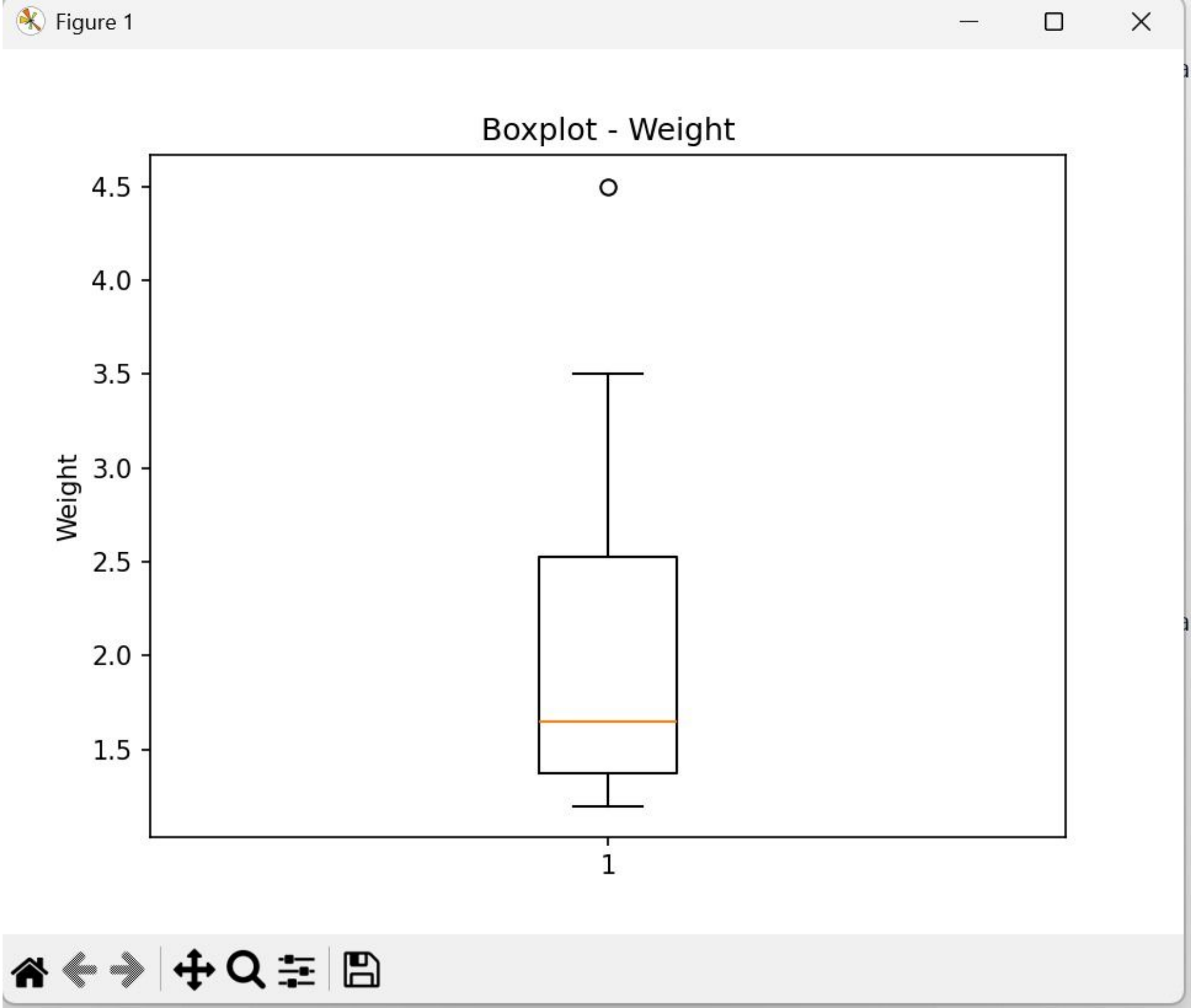
Figure 1
Boxplot - Weight
Weight
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1

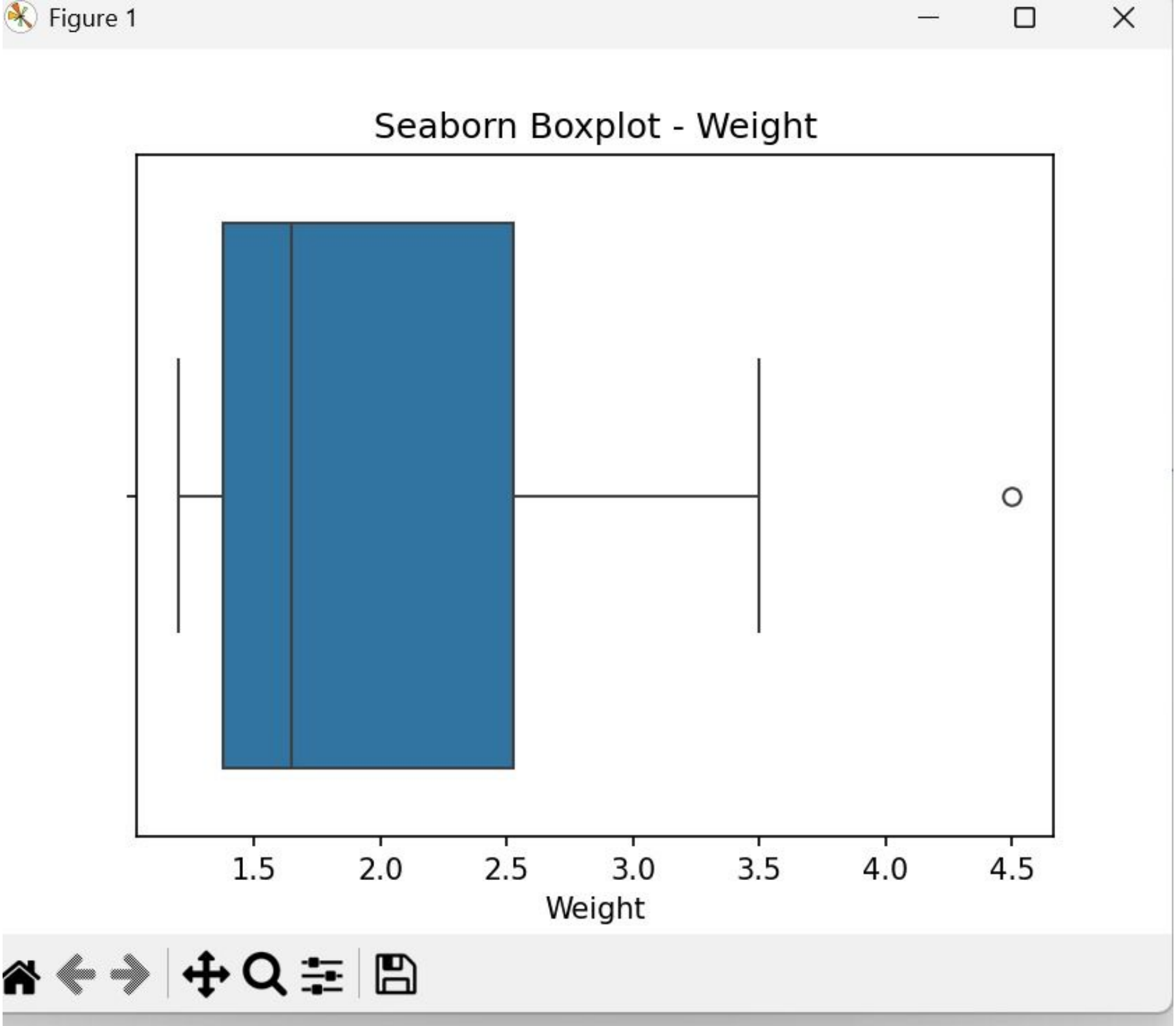
Figure 1
Seaborn Boxplot - Weight
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Weight

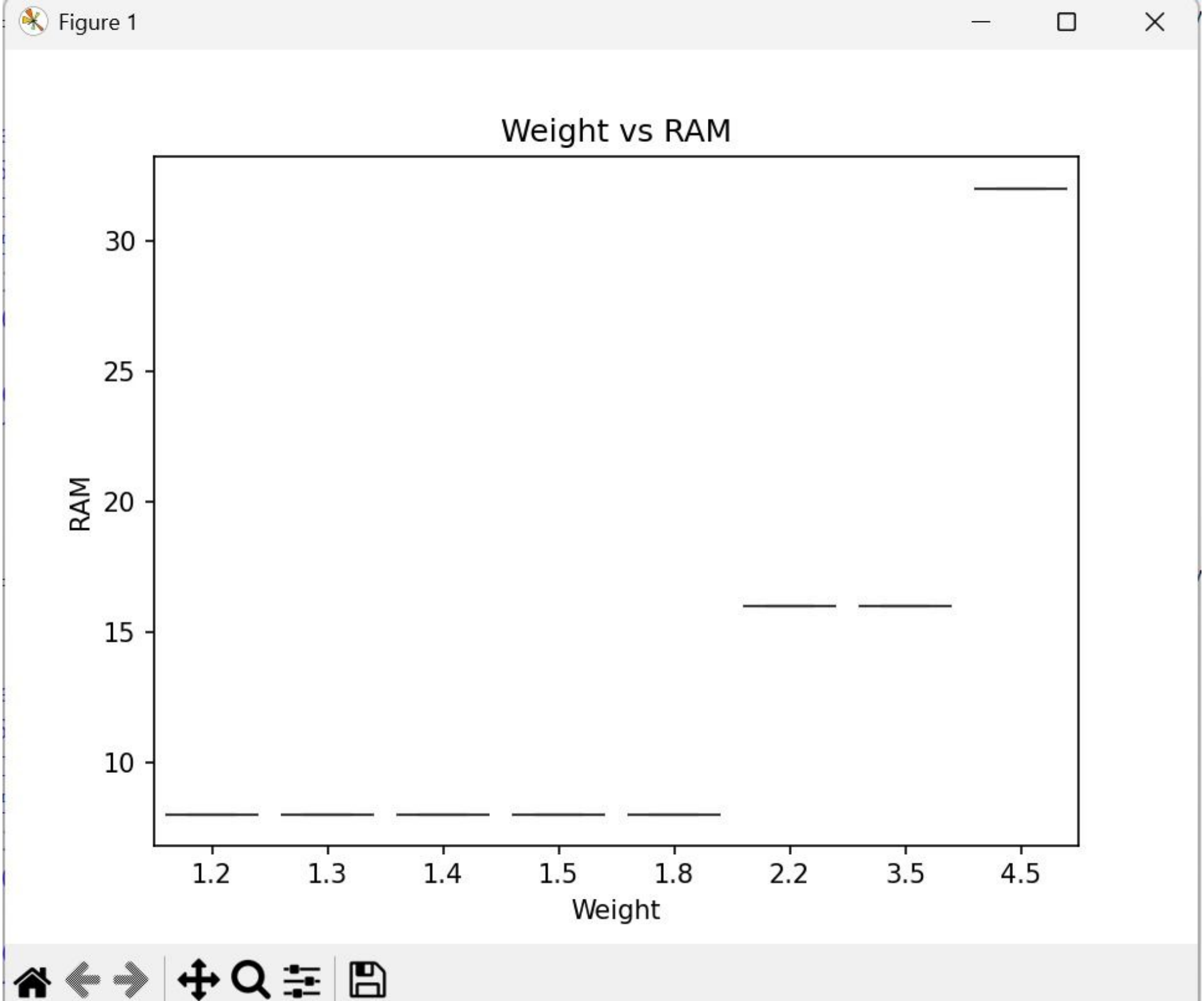
Figure 1
Weight vs RAM
RAM
30
25
20
15
10
1.2
1.3
1.4
1.5
1.8
2.2
3.5
4.5
Weight